Interactive Gantt Chart Builder

by Fraser Davies

Research Report

Submitted in partial fulfilment for the degree of
BSc (Hons) Computer Science

Supervised by Prof. Pierre Le Bras

Heriot-Watt University
School of Mathematical and Computer Sciences

14th November 2025

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Course code and name	F20PA - Honours Project
Type of assessment	Individual
Coursework title	Interactive Gantt Chart Builder
Student name	Fraser Davies
Student ID	H00403114

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Acknowledgements

Acknowledgements text.

Abstract

Abstract text.

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1 Introduction

1.1 Background and Motivation

Project management is critical for students in higher education, particularly those undertaking substantial individual projects such as honours dissertations. Dissertations present unique challenges as they are often the largest and most complex projects students have encountered, requiring sustained effort over extended periods while juggling multiple responsibilities [14]. Effective planning, scheduling, and progress monitoring are essential skills that students must develop to successfully navigate these complex, long-term academic endeavours. However, many students struggle with translating abstract project requirements into concrete, manageable tasks with realistic timelines and dependencies, finding themselves overwhelmed by the scope and duration of dissertation work [14].

Gantt charts have emerged as one of the most widely adopted project management tools, originally developed by Henry Gantt in the early 20th century to visualise project schedules through horizontal bar representations [4, 7]. These visual tools effectively display task durations, dependencies, and timelines, making them invaluable for planning and tracking project progress [5]. In educational contexts, Gantt charts serve as pedagogical tools that help students manage their increasing responsibilities while developing time management skills. Their graphical representation visualises beginning dates, ending dates, and duration parameters, helping to track various projects over specific time periods [3].

Despite their proven utility in professional project management, traditional Gantt chart tools often present usability challenges for students who lack prior project management experience. Many existing solutions are designed for team-based projects and enterprise environments, making them overly complex for individual academic projects. Furthermore, the widespread adoption of visualisation tools in educational settings has been hindered by usability problems and the time required for instructors and students to learn and incorporate these tools into their workflows.

At Heriot-Watt University (HWU), honours project students are currently required to submit project management plans as part of their dissertation deliverables. However, the existing project management system lacks integrated tools that guide students through the process of creating well-structured Gantt charts with appropriate task hierarchies, dependencies, and milestones. This gap in the current system motivated the development of an interactive Gantt chart builder specifically tailored to the needs of honours project students and their supervisors.

1.2 Aim and Objectives

The primary aim of this project is to design, develop, and evaluate an interactive Gantt chart builder that will be integrated into the existing HWU honours project management system.

This tool will provide structured guidance to help students create comprehensive project plans while offering supervisors visibility into student progress and planning quality.

The specific objectives of this project are:

(1) To conduct comprehensive background research on Gantt chart design principles, project management methodologies, and interactive visualisation techniques relevant to educational contexts
(2) To gather and analyse requirements from key stakeholders, including current system developers, supervisors, and students, to ensure the tool meets the needs of its target users
(3) To design an intuitive user interface that guides students through the process of defining milestones, tasks, task hierarchies, and dependencies without requiring prior project management expertise
(4) To develop an interactive Gantt chart visualisation using D3.js (Data-Driven Documents), a powerful JavaScript library that enables the creation of dynamic and interactive data visualisations in web browsers using HTML, SVG, and CSS [6, 18], ensuring seamless integration with the existing HWU project system's technological stack
(5) To implement functionality that allows students and supervisors to explore project timelines, track progress, and reevaluate goals as projects evolve
(6) To provide export capabilities that enable students to include their Gantt charts and associated data in their project documentation
(7) To conduct a rigorous usability evaluation through user studies with students and supervisors to assess the effectiveness of the tool and its guidance features

1.3 Organisation

The organisation of this research report is as follows:

- Section 2: Presents the background research, discussing related work found in the technical literature and its relevance to the project.
- Section 3: ...
- Section 4: ...

2 Background Research

2.1 Gantt Charts and Project Management

Building upon Henry Gantt's original design, modern Gantt charts have evolved to incorporate sophisticated features that support complex project planning. A comprehensive Gantt chart integrates multiple elements: [4, 5]

- A vertical list of tasks or activities on the left side.
- A horizontal timeline indicating project duration (days, weeks, or months)
- Graphical bars representing individual task durations and their temporal placement.
- Lines or arrows indicating dependencies between tasks.
- Labels showing team members or assignees.
- Special markers (typically diamonds) that denote milestones or important project achievements.

Task dependencies are fundamental to effective Gantt chart usage, defining the relationships between tasks and dictating the order in which activities must occur. Project management identifies four primary dependency types [11, 15]. The Finish-to-Start (FS) dependency, where a successor task cannot commence until its predecessor concludes, represents the most common relationship type [10]. Start-to-Start (SS) dependencies allow a task to begin only after another task has also began, enabling parallel workflows that share a start point. Finish-to-Finish (FF) relationships constrain task completion, requiring one task to finish before another can conclude. Finally, the Start-to-Finish (SF) dependency, the least common type, denotes that a task cannot finish until another task begins, typically appearing in handover or transition scenarios.

Beyond dependencies, Gantt charts visualise the critical path the longest sequence of dependent tasks that determines the project's minimum completion time. This enables project managers to distinguish between tasks that are critical to meeting deadlines and those that possess scheduling flexibility [2]. This distinction proves particularly valuable for resource allocation and risk management decisions.

Gantt charts occupy a distinct position within the broader landscape of project management methodologies. Unlike Kanban boards, which emphasise workflow states and work-in-progress limits while focusing on the volume of work completed, in progress, and not yet started, Gantt charts provide explicit temporal representations of task scheduling and dependencies displayed on a visual timeline [13]. Similarly, while Program Evaluation and Review Technique (PERT) charts focus on identifying critical paths through network diagrams using nodes and arrows to represent task dependencies, Gantt charts offer a more intuitive timeline-based bar chart visualisation that facilitates stakeholder communication and progress tracking [8, 12].

2.2 Gantt Charts in Educational Contexts

Within educational contexts, Gantt charts serve multiple pedagogical functions. For educators, Gantt charts provide a structured approach to planning school activities, curriculum development, and lesson planning, with their graphical representation of beginning dates, ending dates, and duration parameters proving particularly effective for tracking various educational activities over specific time periods [3]. Beyond their use by educators, Gantt charts offer valuable support for students undertaking complex academic projects such as dissertations. The ability to visualise project timelines and decompose large, complex undertakings into manageable components with clear deadlines proves particularly valuable when managing extended research projects [14]. This visual representation of project structure enables both students and educators to conceptualise and monitor multiple activities across extended timeframes [3].

However, students often encounter challenges when applying Gantt charts to individual academic projects, particularly concerning the maintenance and updating of project timelines throughout extended research periods. Research with PhD students reveals that while Gantt charts can be beneficial for planning, the unpredictable nature of academic research makes it difficult to maintain detailed charts over multi-year projects, with students reporting frustration when plans deviate from initial timelines [1]. This psychological barrier to updating charts is compounded by the investment required: as one PhD student observes, "if one invest a lot of time and energy in creating a detailed Gantt chart, then one is not gonna be willing to adjust this chart when things didn't work out as planned" [1]. This reluctance to revise detailed plans creates a tension between the effort invested in initial planning and the flexibility required for successful academic project management.

Traditional Gantt chart tools required frequent manual updates to remain accurate, which created a significant maintenance burden. Although modern software reduces this through automation, regular updates are still necessary, and increasing project complexity, particularly in large projects like a dissertation, makes Gantt charts challenging to manage [9]. Modern interactive Gantt chart platforms address many of these challenges through features such as drag-and-drop task scheduling, automatic dependency adjustments, and real-time collaborative updates, which significantly reduce the effort required to maintain project schedules and enable more flexible adaptation to changing project requirements [16]. These interactive capabilities prove particularly valuable for complex projects where tasks and timelines must be frequently adjusted, as they allow users to modify schedules efficiently without the cumbersome manual recalculation required by static chart implementations [16].

2.3 Interactive Data Visualisation

3 Methodology

3.1 Requirement Analysis

4 Feasibility

[17]

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A Project Management

A.1 Gantt Chart

B Generative AI